

South Asia's Water Crisis

The Indus, Ganges-Brahmaputra, and the Himalayan Glacier Emergency

Blue Lotus Research Intelligence | August 2026

Classification: Open Intelligence

Executive Summary

South Asia's water architecture — the Himalayan glaciers, the Indus river system, the Ganges-Brahmaputra basin, and the shared aquifers of the Indo-Gangetic Plain — is under simultaneous stress from converging forces: accelerating glacial retreat driven by climate change, population and economic growth increasing per-capita freshwater demand, and geopolitical competition between India, Pakistan, Afghanistan, and China that makes cooperative water management politically difficult even when hydrologically necessary. India's suspension of the Indus Waters Treaty (IWT) and Afghanistan's plans to build new dams on transboundary rivers have compounded Pakistan's water anxiety at a moment when monsoon extremes are already causing mass evacuations CNA[2025-12-22]. Climate change is making rare flood events more frequent and frequent events more destructive CNA[2025-08-20].

Part I — The Indus Water Treaty: Under Stress from Multiple Directions

The Treaty and Its Suspension

The Indus Waters Treaty, long regarded as one of international water law's success stories for surviving decades of India-Pakistan tensions, has entered a new phase of instability. India suspended the treaty's notification mechanisms — citing cross-border terrorism — a move that the December 2025 Pakistan-Afghanistan border crisis commentary describes as a foundational trigger for Pakistan's compounding water anxieties CNA[2025-12-22]. The suspension, combined with Afghanistan's separate plans to build a dam on the Kunar River and divert water from the Kunar to the Darunta Dam in Nangarhar, threatens to reduce downstream flows into Pakistan from two directions simultaneously CNA[2025-12-22].

Pakistan's declaratory response to Afghanistan's dam plans — that Kabul "cannot disregard Pakistan's water rights" — reflects the existential stakes: Pakistan's agricultural and urban water supply depends on the river systems flowing in from both India and Afghanistan CNA[2025-12-22]. Historically, the Pakistani security establishment had supported the Taliban, and Pakistan had initially welcomed the Taliban's return to power in 2021, but relations deteriorated sharply over the Afghan Taliban's hosting of the Pakistani Taliban (TTP) and over the Kunar River water plans CNA[2025-12-22].

India's Regional Water Diplomacy

India's water posture is not purely adversarial. India funded and completed the Salma Dam (also known as the India-Afghanistan Friendship Dam) on the Hari River in Herat province, contributing around

US\$300 million. The dam generates 42 MW of electricity and irrigates approximately 75,000 hectares of Afghan farmland — a project that simultaneously deepened India-Afghanistan ties and reduced Pakistan's regional influence CNA[2025-12-22].

The practical consequence of simultaneous IWT suspension and Afghan dam construction is that Pakistan faces upstream pressure from both neighbours at a moment when its own monsoon resilience is deteriorating. In August 2025, Pakistan evacuated tens of thousands of people from low-lying areas along the Sutlej River in Punjab province after India released water from swollen rivers — a vivid illustration of how upstream water management decisions translate directly into downstream humanitarian crises CNA[2025-08-26].

Part II — Monsoon Extremes: Floods as the Near-Term Crisis

Pakistan's Monsoon Vulnerability

The monsoon season brings South Asia approximately three-quarters of its annual rainfall, vital for agriculture and food security, but it also brings destruction CNA[2025-08-16]. In August 2025, monsoon rains killed over 340 people in Pakistan, with the provincial government declaring the mountainous districts of Buner, Bajaur, Swat, Shangla, Mansehra, and Battagram as disaster-hit areas CNA[2025-08-16]. Eleven more were killed in Pakistan-administered Kashmir CNA[2025-08-16].

Climate scientists have directly linked this pattern to warming: in mountain regions, warming accelerates glacier melt, adds excessive moisture to the atmosphere, and destabilises mountain slopes, making rare events more frequent and frequent events more destructive CNA[2025-08-20]. Cloudbursts — sudden intense rainfall events — are becoming a signature hazard of South Asia's changing climate CNA[2025-08-20].

India and Bangladesh: The Brahmaputra Flood Corridor

In June 2022, massive flooding and mudslides in India's northeastern Assam state killed at least 62 people, with 32 of the state's 35 districts submerged. The Brahmaputra River — one of Asia's largest — overflowed its banks, causing one of the most severe flooding events the state had seen in years CNA[2022-06-19]. Bangladesh, downstream of both the Ganges and Brahmaputra systems, faces an additional threat: river erosion driven by the swelling Padma (a major tributary of the Ganges) is consuming up to 5 metres of riverbank in a single day. More than 9,000 people were displaced by Padma-driven erosion alone in just five years, a process that local experts and government officials describe as being rapidly accelerated by climate change CNA[2018-10-04].

Part III — The Ganges: Sacred River, Stressed Ecosystem

Pollution and the Modi Clean-Up Programme

India's holy Ganges begins as a crystal-clear river high in the Himalayas but is transformed into what reporters describe as toxic sludge on its journey through cities, industrial hubs, and past millions of devotees CNA[2017-07-10]. Industrial waste and sewage pour in from open drains; at some stretches the river turns red from tannery effluent, with chemical-soaked hides dumped directly into the current CNA[2017-07-10]. Prime Minister Modi's government pledged a US\$3 billion clean-up plan, committing to build more treatment plants and relocate over 400 tanneries away from the river — but as of reporting, less than a quarter of an estimated 4,800 miles of sewage infrastructure had been treated, and the plan was described as badly behind schedule CNA[2017-07-10].

The Ganges remains the spiritual heart of Hindu India. The Maha Kumbh Mela festival — held once every 12 years at the confluence of the Ganges, Yamuna, and mythical Saraswati rivers in Prayagraj — drew

400 million pilgrims in January 2025, the largest religious gathering in human history CNA[2025-01-13]. The coexistence of pilgrimage at this scale with acute river pollution defines the contradiction at the centre of India's Ganges governance challenge.

Part IV — Himalayan Glacier Retreat: The Underlying Driver

Glacial Melt and the Chamoli Disaster

China's glaciers on the Qinghai-Tibet Plateau have shrunk by approximately 15 per cent — or 8,000 square kilometres — over the past 30 years as a result of climate change, according to the Chinese Academy of Sciences CNA[2014-05-24]. The Tibetan Plateau is the headwater source for most of South Asia's major rivers; glacial shrinkage there directly affects long-term river base flows across the entire region.

The most dramatic recent illustration of glacial hazard in the Indian Himalayas was the Chamoli disaster of February 7, 2021, in Uttarakhand. When scientists and observers saw the devastating deluge of water and debris crash downstream from a Himalayan glacier, they immediately identified it as the realisation of risks that glaciologists had long warned about CNA[2021-02-10]. Climate change, by warming mountain regions and destabilising glacial and permafrost systems, is systematically increasing the probability of such catastrophic events CNA[2025-08-20].

Part V — Groundwater Depletion: India's Punjab Crisis

The "Granary" Running Dry

India's northwestern state of Punjab, hailed as the country's granary, faces a drastic decline in agricultural output unless it halts the rapid depletion of its groundwater. Groundwater irrigates almost three-quarters of Punjab's agricultural land, but groundwater levels are dropping by 40 to 50 centimetres per year CNA[2016-10-17]. The state relies on nearly 1.4 million tube wells to extract groundwater, which exacerbates the loss of more accessible surface water CNA[2016-10-17].

The scale of Punjab's agricultural dependency makes this a national-level risk: despite accounting for only 1.5 per cent of India's geographical area, Punjab has contributed approximately 35 per cent of the nation's rice production and 60 per cent of its wheat over the past two decades CNA[2016-10-17]. Rice and wheat make up 81 per cent of Punjab's irrigated crops — both are water-intensive, and experts warn that the state's agricultural success may not be sustainable at current extraction rates CNA[2016-10-17]. Around 3.5 million of Punjab's 9.1 million workers make a living from agriculture or associated activities, meaning groundwater depletion is not merely an environmental issue but a direct livelihood threat at scale CNA[2016-10-17].

Conclusion: Water as the Region's Existential Variable

South Asia's water crisis is not a future scenario — it is an active, multi-front emergency. India's suspension of IWT mechanisms, Afghanistan's planned Kunar River diversions, and Pakistan's mass evacuations during the 2025 monsoon season demonstrate that water has already become a lever of geopolitical pressure CNA[2025-12-22] CNA[2025-08-26]. Himalayan glacier retreat is the underlying structural driver, progressively reducing the long-term reliability of river base flows while simultaneously increasing the frequency and severity of glacial flood events CNA[2021-02-10] CNA[2025-08-20]. In the Indo-Gangetic Plain, groundwater depletion in India's most productive agricultural state is advancing at a rate that experts describe as unsustainable CNA[2016-10-17].

The diplomatic architecture — a stressed Indus Waters Treaty, no binding water agreement between India and China on the Brahmaputra, and no multilateral South Asian water governance mechanism — remains insufficient for the governance challenge ahead. Without cooperative institutions, water conflicts will be managed through unilateral dam construction, upstream flow manipulation, and periodic humanitarian crisis rather than shared management.

Blue Lotus Research Intelligence | South Asia Series | Water Crisis | August 2026

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